

4

MEASUREMENT AND SPACE

TRIGONOMETRY

In the second century BCE, the Greek astronomer Hipparchus could calculate distances to the moon and the Sun and he was the first scientist to chart the positions of over 850 stars. How was he able to achieve this over 2100 years ago? Hipparchus started a branch of mathematics called **trigonometry**, meaning 'triangle measure', which uses angles, triangles and circles to calculate lengths and distances that cannot be physically measured. Trigonometry is used widely today in engineering, surveying, navigation, astronomy, electronics and construction.



Chapter outline

	Working mathematically				
4.01 The sides of a right-angled triangle	U	F			C
4.02 The trigonometric ratios	U	F		R	C
4.03 Similar right-angled triangles	U	F		R	C
4.04 Trigonometry on a calculator	U	F			
4.05 Finding an unknown side	U	F	PS		C
4.06 Finding more unknown sides	U	F	PS		
4.07 Finding an unknown angle	U	F	PS		

Note: Bearings and angles of elevation and depression will be covered in Year 10 (or Year 9 Advanced),

Wordbank

adjacent side In a right-angled triangle, the side that is next to a given angle and pointing to the right angle

hypotenuse The longest side of a right-angled triangle, the side opposite the right angle

minute (′) A unit for measuring angle size, $\frac{1}{60}$ of a degree

opposite side In a right-angled triangle, the side that is facing a given angle and not one of its arms

theta (θ) A letter of the Greek alphabet used as a variable for angles

trigonometric ratio The ratio of 2 sides in a right-angled triangle, for example, sine is the ratio of the opposite side to the hypotenuse

Quiz
Wordbank 4

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (8):

4.01 The sides of a right-angled triangle
4.02 The trigonometric ratios
4.04 Trigonometry on a calculator
• Rounding angles
4.05 Finding an unknown side
• Trigonometry
4.06 Finding the hypotenuse
4.07 Finding an unknown angle
• Trigonometry

Twig videos (3):

SkillCheck Measuring the Earth • The tunnel of Samos • Distance to the Sun and Moon

Quizzes (5):

- Wordbank 4
- SkillCheck 4
- Mental skills 4
- Language of maths 4
- Test yourself 4

Worksheets (6):

4.03 Investigating the tangent ratio
4.04 Trigonometric calculations
4.06 Finding an unknown side
4.07 Finding an unknown angle
• Trigonometry problems

Mind map Trigonometry

Puzzles (6):

4.02 Trigonometry match-up
4.04 Trigonometry squaresaw
4.06 Trigonometry equations 1
4.07 Trigonometry squaresaw
• Trigonometry: Finding angles
• Trigonometry equations 2

Language of maths Trigonometry crossword

Technology (1):

4.02 Trigonometry values

Spreadsheet (1):

4.02 Trigonometry values

Presentation (1):

4.07 Applications of triangles

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- investigate the sine, cosine and tangent ratios in similar right-angled triangles
- use trigonometry to solve problems involving right-angled triangles
- find unknown sides and angles in right-angled triangles, where the angle is measured in degrees and minutes

Quiz
SkillCheck 4

Videos
Measuring the Earth
The tunnel of Samos
Distance to the Sun and Moon

SkillCheck ANSWERS ON P. 570

1 Simplify each fraction.

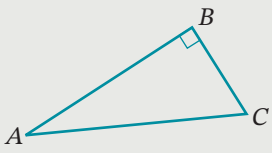
a $\frac{15}{25}$ **b** $\frac{9}{12}$ **c** $\frac{4}{10}$

2 Convert each fraction to a decimal, correct to 3 decimal places.

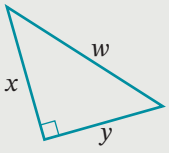
a $\frac{3}{8}$ **b** $\frac{5}{7}$ **c** $\frac{2}{9}$

3 For each triangle, name the hypotenuse.

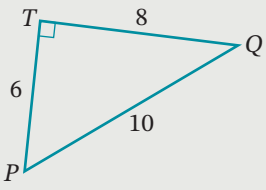
a



b



c

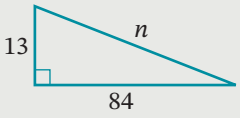


4 Solve each equation.

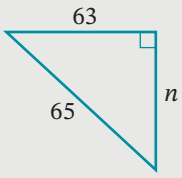
a $\frac{x}{5}=7$ **b** $\frac{h}{4}=8.3$ **c** $\frac{45}{y}=9$

5 For each triangle, find the value of n.

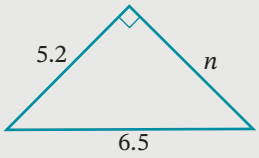
a



b



c



6 Round each time to the nearest hour.

a 8 h 18 min **b** 3 h 45 min **c** 1 h 30 min

7 Convert each time to hours and minutes.

a 4.7 h **b** 2.25 h **c** 6.85 h

The sides of a right-angled triangle 4.01

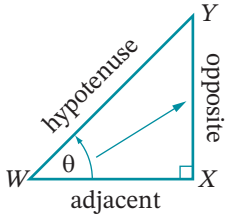
The 3 sides of a **right-angled triangle** have special names. These names depend on the position of a given angle in the triangle.

- The **hypotenuse** is the longest side and is always opposite the right angle
- The **opposite side** directly faces the given angle
- The **adjacent side** runs from the given angle to the right angle

We have already learnt about the hypotenuse in Pythagoras' theorem.

Adjacent means 'next to'.

In this diagram, the marked angle $\angle W$ has also been labelled with the Greek letter, θ, ('theta').



The hypotenuse is WY.

The opposite side is XY.

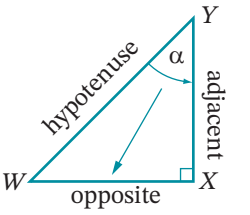
The adjacent side is WX.

In this second diagram, the marked angle $\angle Y$ has also been labelled with the Greek letter α ('alpha'), so the opposite and adjacent sides are different.

The hypotenuse is WY.

The opposite side is WX.

The adjacent side is XY.

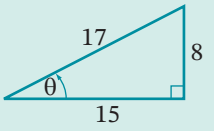


Video
The sides of a right-angled triangle

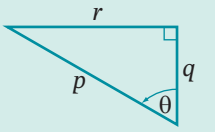
Example 1

For each triangle, name the hypotenuse, opposite and adjacent sides for angle θ.

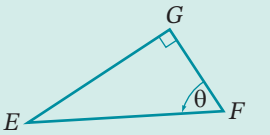
a



b



c



Solution

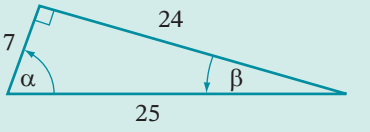
a Hypotenuse is 17.
Opposite side is 8.
Adjacent side is 15.

b Hypotenuse is p
Opposite side is r.
Adjacent side is q.

c Hypotenuse is EF.
Opposite side is EG.
Adjacent side is FG.

Example 2

For angles α and β, name the adjacent side.



Solution

For α, the adjacent side is 7.

For β, the adjacent side is 24.

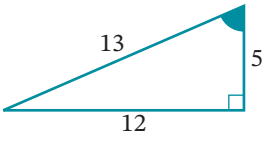
EXERCISE 4.01 ANSWERS ON P. 570

The sides of a right-angled triangle U F C

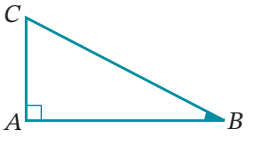
EXAMPLE 1

1 For the marked angle in each triangle, name the hypotenuse, opposite and adjacent sides. C

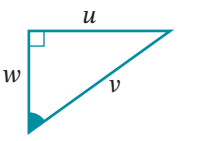
a



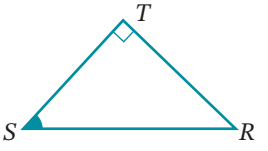
b



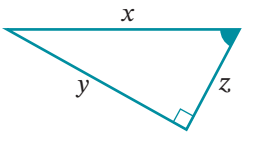
c



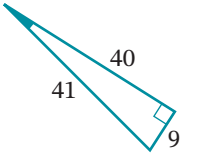
d



e



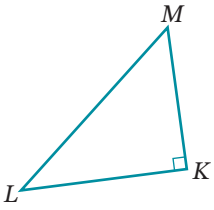
f



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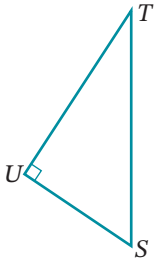
2 For $\triangle LKM$, name the angle:



- **a** opposite the hypotenuse
- **b** opposite side KM
- **c** opposite side LK
- **d** adjacent to side KM
- **e** adjacent to side LK.

3 Which one of these statements is **false** about $\triangle TSU$?

Select the correct answer **A**, **B**, **C** or **D**.

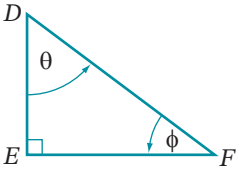


- **A** The adjacent side to $\angle S$ is US.
- **B** The adjacent side to $\angle T$ is TU.
- **C** The hypotenuse is UT.
- **D** The opposite side to $\angle T$ is US.

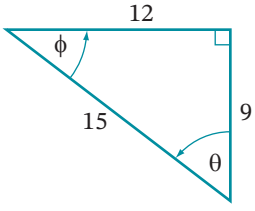
EXAMPLE 2

4 For each triangle, find the opposite sides for angles θ ('theta') and ϕ ('phi').

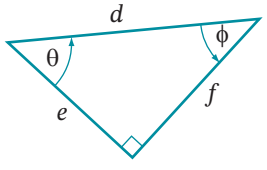
a



b

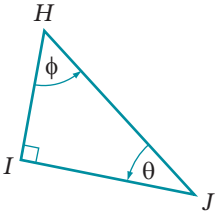


c

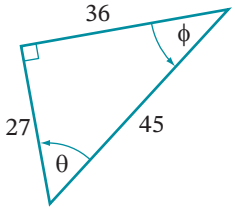


5 For each triangle, find the adjacent sides for angles θ and ϕ.

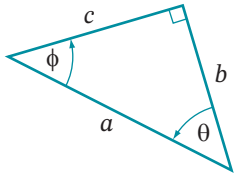
a



b



c



6 Which side of a right-angled triangle is fixed and does not depend on a given angle?

Select **A**, **B**, **C** or **D**.

A adjacent **B** hypotenuse **C** opposite **D** shortest

7 Given each description of a right-angled triangle, sketch the triangle with the correctly labelled vertices and angle. C

- **a** $\triangle ABC$ has hypotenuse AB and side AC opposite angle θ
- **b** $\triangle XYZ$ has hypotenuse YZ and side XZ adjacent to angle α
- **c** $\triangle PRQ$ has side RQ opposite $\angle P$ and adjacent to $\angle R$
- **d** $\triangle DEF$ is right-angled at E, with the opposite and adjacent sides of $\angle D$ equal

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Did you know?

The Greek alpha-bet

Here are 8 letters (in lower-case and capitals) from the Greek alphabet:

α, A alpha	β, B beta	γ, Γ gamma	δ, Δ delta
θ, Θ theta	π, Π pi	σ, Σ sigma	ω, Ω omega

Greek letters are commonly used as variables, particularly in geometry and trigonometry.

1 **Find how many letters there are in the Greek alphabet, and name each one.**

2 **Compare the Greek alphabet with our Roman alphabet.**

3 **Can you see where the word alphabet comes from? Explain how it originated.**

4.02 The trigonometric ratios

Puzzle
Trigonometry match-up

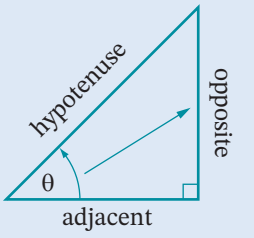
Technology
Trigonometry values

Spreadsheet
Trigonometry values

There are 3 special fractions called **trigonometric ratios** that relate the lengths of 2 sides of a right-angled triangle: **sine**, **cosine** and **tangent**.

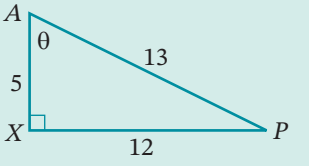
The trigonometric ratios

Ratio	Abbreviation	Meaning
sine	sin	$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$
cosine	cos	$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$
tangent	tan	$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$



Example 3

In $\triangle AXP$, find $\sin\theta$, $\cos\theta$ and $\tan\theta$.



Solution

For angle θ, opposite = 12, adjacent = 5, hypotenuse = 13.

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{12}{13} \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{5}{13}$$

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{12}{5}$$

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A useful **mnemonic** (memory aid) for remembering the 3 ratios is to look at the initials of the words in the ratios:

$$\sin = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{O}{H} \quad \text{'SOH'}$$

$$\cos = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{A}{H} \quad \text{'CAH'}$$

$$\tan = \frac{\text{opposite}}{\text{adjacent}} = \frac{O}{A} \quad \text{'TOA'}$$

If you remember **SOH-CAH-TOA** (pronounced 'so-car-toe-ah'), then you can remember the ratios for sin, cos and tan. Some students also learn a phrase where the first letter of each word follows the SOH-CAH-TOA sequence, for example, '**S**un **O**ver **H**ead **C**aused **A** **H**uge **T**an **O**n **A**rms'. Find your own mnemonic for the 3 ratios.

4.02

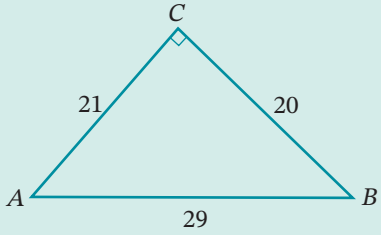
Video The trigonometric ratios

Example 4

For the triangle shown, find:

a $\sin A$ **b** $\cos B$

c $\tan A$ **d** $\sin B$



Solution

For angle A, opposite = 20, adjacent = 21, hypotenuse = 29

For angle B, opposite = 21, adjacent = 20, hypotenuse = 29

a $\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{20}{29}$

b $\cos B = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{20}{29}$

c $\tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{20}{21}$

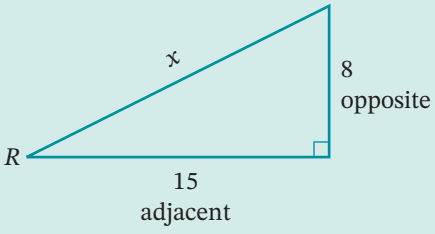
d $\sin B = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{21}{29}$

Example 5

If $\tan R = \frac{8}{15}$, find the value of $\sin R$ and $\cos R$.

Solution

Since $\tan R = \frac{\text{opposite}}{\text{adjacent}} = \frac{8}{15}$, draw a right-angled triangle that has an angle R with opposite side 8 and adjacent side 15. Let x be the length of the hypotenuse.



Find x using Pythagoras' theorem.

$$x^2 = 8^2 + 15^2$$
$$= 289$$
$$x = \sqrt{289}$$
$$= 17$$

$$\therefore \sin R = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{8}{17}$$

$$\cos R = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{15}{17}$$

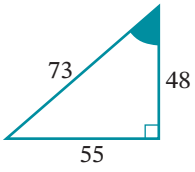
EXERCISE 4.02 ANSWERS ON P. 570

The trigonometric ratios U F R C

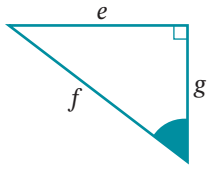
EXAMPLE 3

1 For each marked angle, find the sine, cosine and tangent ratios. C

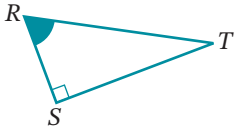
a



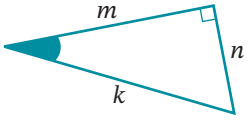
b



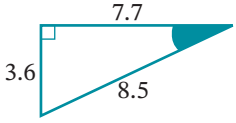
c



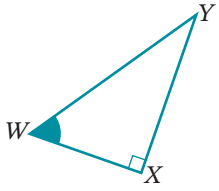
d



e



f

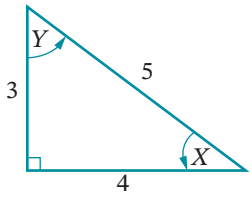


EXAMPLE 4

2 For the triangle shown, find:

a $\cos X$ **b** $\tan Y$

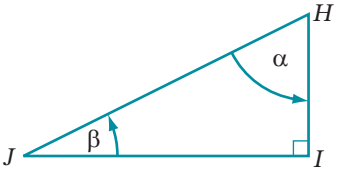
c $\sin X$ **d** $\sin Y$



3 Complete each statement below with the correct angle (α or β).

a $\sin\square = \frac{IJ}{HJ}$ **b** $\sin\square = \frac{HI}{HJ}$ **c** $\cos\square = \frac{IJ}{HJ}$

d $\cos\square = \frac{HI}{HJ}$ **e** $\tan\square = \frac{IJ}{HI}$ **f** $\tan\square = \frac{HI}{IJ}$



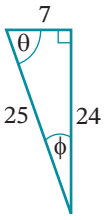
Foundation Standard Complex

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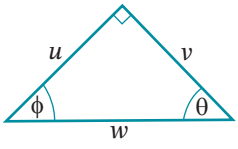
4 For each triangle below, find:

i $\tan\theta$ **ii** $\cos\theta$ **iii** $\cos\phi$ **iv** $\tan\phi$ ϕ = 'phi'

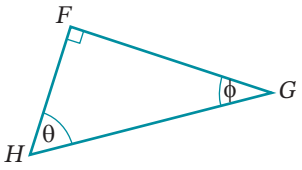
a



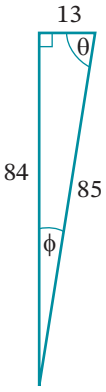
b



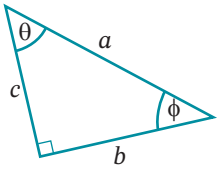
c



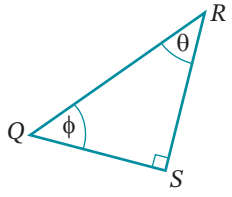
d



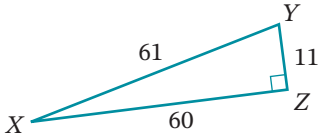
e



f



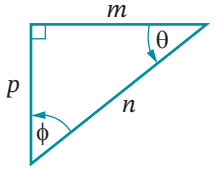
5 For each fraction, write a correct trigonometric ratio involving angle X or Y in the triangle. C



a $\frac{60}{11}$ **b** $\frac{11}{60}$

c $\frac{11}{61}$ **d** $\frac{60}{61}$

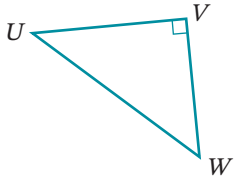
6 Which ratio is equal to $\frac{p}{m}$? Select the correct answer **A**, **B**, **C** or **D**.



A $\cos\theta$ **B** $\cos\phi$

C $\tan\theta$ **D** $\tan\phi$

7 Which statement is true for this triangle? Select **A**, **B**, **C** or **D**. R C



A $\sin U = \cos W$ **B** $\tan U = \sin W$

C $\cos U = \tan W$ **D** $\tan U = \tan W$

8 Sketch a right-angled triangle for each trigonometric ratio, then use Pythagoras' theorem to find the length of the unknown side and the other 2 trigonometric ratios for the same angle.

a $\tan A = \frac{5}{12}$ **b** $\sin B = \frac{3}{5}$ **c** $\cos X = \frac{9}{41}$ **d** $\sin Y = \frac{7}{25}$

4.02

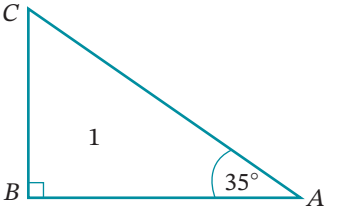
EXAMPLE 5

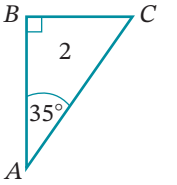
Foundation Standard Complex

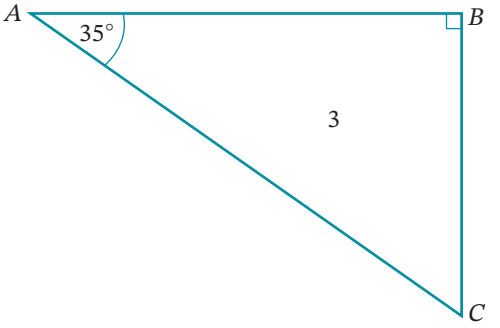
4.03 Similar right-angled triangles

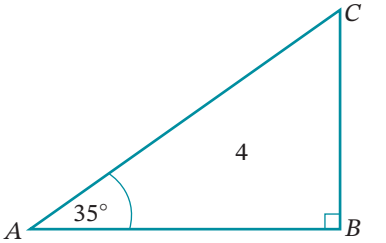
Worksheet
Investigating the tangent ratio

In each right-angled triangle below, $\angle A = 35°$.









Furthermore, because $\angle B = 90°$, $\angle C = 180° - 90° - 35° = 55°$ because of the angle sum of a triangle.

These 4 triangles are called **similar triangles** because they have the same shape but are not the same size. In fact, they are enlargements or reductions of one another.

Example 6

For each triangle above, measure the length of each side (correct to the nearest mm) and then calculate sin A, cos A and tan A as decimals (correct to 2 decimal places).

Solution

	Side length (mm)			Trigonometric ratio		
	BC (opp)	AB (adj)	AC (hyp)	$\sin A = \frac{BC}{AC}$	$\cos A = \frac{AB}{AC}$	$\tan A = \frac{BC}{AB}$
1	25	36	44	$25 \div 44 \approx 0.57$	$36 \div 44 \approx 0.82$	$25 \div 36 \approx 0.69$
2	14	20	24	0.58	0.83	0.70
3	39	57	69	0.57	0.83	0.68
4	29	41	50	0.58	0.82	0.71

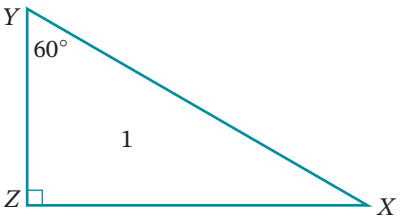
Note: See **Technology: Similar right-angled triangles** after Exercise 4.03 for a dynamic geometry activity based on this example.

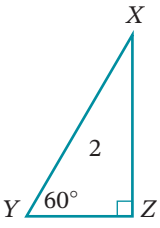
Similar triangles have matching equal angles. We will learn more about them in Chapter 13, *Similar figures*.

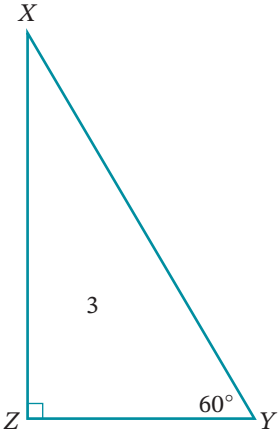
Similar right-angled triangles U F R C

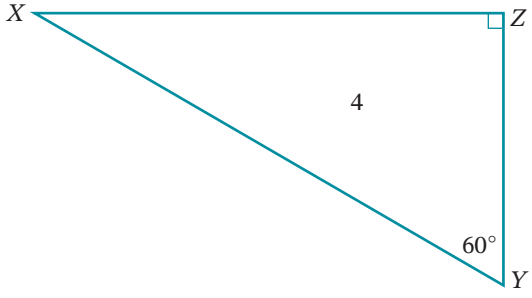
EXAMPLE 6

1 a For each similar right-angled triangle below, measure the length of each side (correct to the nearest mm) and then calculate sin Y, cos Y and tan Y as decimals (correct to 2 decimal places). Copy and complete the table below. R C









	Side length (mm)			Trigonometric ratio		
	XZ (opp)	ZY (adj)	XY (hyp)	$\sin Y = \frac{XZ}{XY}$	$\cos Y = \frac{ZY}{XY}$	$\tan Y = \frac{XZ}{ZY}$
1						
2						
3						
4						

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b What do you notice about the value of sin Y for all 4 similar right-angled triangles?

c Use your calculator to evaluate sin 60° by pressing sin 60 =. What do you notice about your answer?

d What do you notice about the value of cos Y for all 4 similar triangles?

e Use your calculator to evaluate cos 60°. What do you notice about your answer?

f What do you notice about the value of tan Y for all 4 similar triangles?

g Use your calculator to evaluate tan 60°. What do you notice about your answer?

2 **a** Draw 4 similar right-angled triangles that have an angle of 48°, measure the length of each side (correct to the nearest mm) and then calculate sin 48°, cos 48° and tan 48° as decimals (correct to 2 decimal places). Copy and complete the table. R C

	Side length (mm)			Trigonometric ratio		
	Opp	Adj	Hyp	$\sin 48° = \frac{\text{opp}}{\text{hyp}}$	$\cos 48° = \frac{\text{adj}}{\text{hyp}}$	$\tan 48° = \frac{\text{opp}}{\text{adj}}$
1						
2						
3						
4						

b Examine the value of sin 48° for all 4 similar triangles, then evaluate sin 48° on a calculator. What do you notice?

c Examine the values of cos 48°, then evaluate cos 48° on a calculator.

d Examine the values of tan 48°, then evaluate tan 48° on a calculator.

3 For each trigonometric ratio, draw a large right-angled triangle with the given angle, then by measurement and calculation, find the value of the ratio, correct to 3 decimal places. Compare your answer to the calculator's answer. R C

a tan 55° **b** cos 39° **c** sin 67° **d** cos 21°

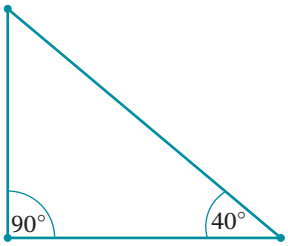
Technology

Similar right-angled triangles

In this activity, use dynamic geometry software to measure and calculate trigonometric ratios.

1 **a** Before you start, set the measuring tool to measure angles to degrees.

b By constructing intervals, draw 2 similar right-angled triangles. Then use the measuring tool to measure the right angle and another angle (as shown right).

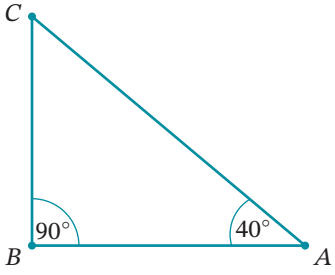


c Now keeping the triangle right-angled, move any of the other 2 vertices so that one angle is 40°.

2 Draw 2 more similar right-angled triangles with a 40° angle.

3 Label the vertices of each triangle as A, B and C.

Make sure $\angle A = 40°$ and $\angle B = 90°$.



4 **a** Copy this table.

	Side length (mm)			Trigonometric ratio		
	BC (opp)	AB (adj)	AC (hyp)	$\sin A = \frac{BC}{AC}$	$\cos A = \frac{AB}{AC}$	$\tan A = \frac{BC}{AB}$
1						
2						
3						

b Use the measurement tool to measure the sides of each triangle and write them in the table.

c Then, for each triangle, calculate the trigonometric ratios, correct to 2 decimal places.

d What do you notice about the values of each ratio for all of the triangles?

4.04 Trigonometry on a calculator

Worksheet
Trigonometric calculations

Puzzles
Trigonometry squaresaw

In the previous section, we discovered that for any particular angle, the sine, cosine and tangent ratios stay constant (the same) for all right-angled triangles with that angle. For example, $\sin 32° \approx 0.53$ always, no matter what size the similar right-angled triangle is.

Constancy of the sine, cosine and tangent ratios

For any given angle, the values of the sine, cosine and tangent ratios are constant.

This means that the value of a trigonometric ratio can be easily found on a calculator rather than through constructing and measuring triangles.

Angles are measured in **degrees**, but one degree can be subdivided into 60 **minutes**.

One minute can be further subdivided into 60 **seconds**. The abbreviations for minutes and seconds are shown below.

Degrees and minutes

$1° = 60'$ (1 degree = 60 minutes)

$1' = 60''$ (1 minute = 60 seconds)

For example, an angle size of $48° 35' 56''$ is 48 degrees, 35 minutes and 56 seconds, about halfway between 48° and 49°.

To enter degrees, minutes and seconds into a scientific calculator, use the °''' or DMS (Degrees-Minutes-Seconds) key.

Videos
Trigonometry on a calculator

Example 7

Evaluate each expression correct to 2 decimal places.

a $\sin 46°$ **b** $\tan 57.4°$ **c** $4 \cos 20°$

d $68.3 \sin 38°25'$ **e** $\dfrac{23}{\cos 18°50'}$

Solution

Make sure that your calculator is in the degrees mode (D or DEG) or your answer will be incorrect.

a $\sin 46° = 0.71933...$
≈ 0.72

On calculator: sin 46 =

b $\tan 57.4° = 1.56365...$

This angle is 57.4° (a decimal), not 57°4′.

≈ 1.56

On calculator: tan 57.4 =

c $4 \cos 20° = 3.75877...$

This means $4 \times \cos 20°$.

≈ 3.76

On calculator: 4 cos 20 =

d $68.3 \sin 38°25′ = 42.43996...$

≈ 42.44

On calculator: 68.3 sin 38 °’’’ 25 °’’’ =

e $\dfrac{23}{\cos 18°50′} = 24.30103...$

≈ 24.30

On calculator: 23 ÷ cos 18 °’’’ 50 °’’’ =

4.04

Rounding angle sizes

When rounding an angle to the nearest degree or minute, use 30 as the halfway mark.

Example 8

Round each angle correct to the nearest degree.

a 73°27′ **b** 9°41′ **c** 42°30′ **d** 128°29′47″

Video Rounding angles

Solution

a $73°27′ \approx 73°$ — $27′ < 30′$, so round down

b $9°41′ \approx 10°$ — $41′ \geq 30′$, so round up

c $42°30′ \approx 43°$ — $30′ \geq 30′$, so round up

d $128°29′ 47″ \approx 128°$ — $29′ < 30′$, so round down

Example 9

Round each angle correct to the nearest minute.

a 33°53′30″ **b** 44°15′40″

Video Rounding angles

Solution

a $33°53′30″ \approx 33°54″$ — $30″ \geq 30″$, so round up

b $44°15′40″ \approx 44°16″$ — $40″ \geq 30″$, so round up

Example 10

Convert each angle size to degrees and minutes, correct to the nearest minute.

a 82.5° **b** 60.81°

Solution

a $82.5° = 82°30'$ On a calculator: 82.5 = °’’’

b $60.81° = 60°48'36''$
$\approx 60°49'$ On a calculator: 60.81 = °’’’

EXERCISE 4.04 ANSWERS ON P. 571

Trigonometry on a calculator U F

EXAMPLE 7

1 Evaluate each expression correct to 2 decimal places.

a	$\tan 84°$	**b**	$\cos 15°$	**c**	$\tan 47°$	**d**	$\sin 33°$
e	$\sin 77°$	**f**	$\cos 60.1°$	**g**	$\tan 39.55°$	**h**	$\cos 18°$
i	$8 \tan 75°$	**j**	$14 \sin 56°$	**k**	$12 \div \tan 20°$	**l**	$\frac{7}{\sin 43°}$
m	$50 \times \sin 70° 34'$	**n**	$66.2 \cos 81° 42'$	**o**	$18.53 \sin 11° 8'$	**p**	$\frac{27}{\cos 35°22'}$
q	$\frac{44.5}{\tan 65°58'}$	**r**	$\frac{200}{\sin 54°47'}$	**s**	$24.1 \div \tan 63° 2'$	**t**	$\frac{15.7}{\cos 21°8'}$

EXAMPLE 8

2 Round each angle size correct to the nearest degree.

a	$27°54'$	**b**	$40°30'$	**c**	$19°18'$	**d**	$33°7'25''$
e	$33°41'5''$	**f**	$56.4°$	**g**	$29.75°$	**h**	$44°18'$

EXAMPLE 9

3 Round each angle size correct to the nearest minute.

a	$68°39'42''$	**b**	$54°22'21''$	**c**	$24°46'30''$	**d**	$18°30'27''$
e	$9°10'55''$	**f**	$47°59'9.5''$	**g**	$3°45'35''$	**h**	$57°10'29''$

EXAMPLE 10

4 Convert each angle size to degrees and minutes, correct to the nearest minute.

a	$55.5°$	**b**	$14.15°$	**c**	$72.38°$	**d**	$33.77°$
e	$66.41°$	**f**	$7.875°$	**g**	$28.123°$	**h**	$31.046°$
i	$34.45°$	**j**	$71.087°$	**k**	$5.4829°$	**l**	$69.4545°$

5 By guess-and-checking with your calculator, find the angle size, θ (to the nearest degree), that gives each value.

a	$\sin\theta = 0.7880$	**b**	$\tan\theta = 0.2493$	**c**	$\tan\theta = 1.2799$
d	$\cos\theta = 0.5$	**e**	$\sin\theta = 0.5446$	**f**	$\cos\theta = 0.8829$
g	$\tan\theta = 0.7265$	**h**	$\sin\theta = 0.9998$		

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Mental skills 4: Maths without calculators ANSWERS ON P. 571

Quiz
Mental skills 4

Estimating answers

A quick way of estimating an answer is to round each number in the calculation.

1 Study each example.

a $631 + 280 + 51 + 43 + 96 \approx 600 + 300 + 50 + 40 + 100$
$= (600 + 300 + 100) + (50 + 40)$
$= 1000 + 90$
$= 1090$ (Actual answer = 1101)

b $55 + 132 - 34 + 17 - 78 \approx 60 + 130 - 30 + 20 - 80$
$= (60 + 20 - 80) + (130 - 30)$
$= 0 + 100$
$= 100$ (Actual answer = 92)

c $78 \times 7 \approx 80 \times 7$
$= 560$ (Actual answer = 546)

d $510 \div 24 \approx 500 \div 20$
$= 50 \div 2$
$= 25$ (Actual answer = 21.25)

2 Now estimate each answer.

a $27 + 11 + 87 + 142 + 64$
b $55 + 34 - 22 - 46 + 136$
c $684 + 903$
d $35 + 81 + 110 + 22 + 7$
e $517 - 96$
f $210 - 38 - 71 + 151 - 49$
g $766 - 353$
h 367×2
i 83×81
j 984×16
k $828 \div 3$
l $507 \div 7$

3 Study each example involving decimals.

a $20.91 - 11.3 + 2.5 \approx 21 - 11 + 3$
$= 13$ (Exact answer = 12.11)

b $4.78 \times 19.2 \approx 5 \times 20$
$= 100$ (Exact answer = 91.776)

c $75.13 \div 8.4 \approx 75 \div 8$
$< 80 \div 8$
< 10
≈ 9 (Exact answer = 8.944...)

d $\dfrac{37.6+9.3}{41.2-12.7} \approx \dfrac{38+9}{40-13}$

$= \dfrac{47}{27}$

$\approx \dfrac{50}{30}$

≈ 1.6 (Exact answer = 1.645...)

4 Now estimate each answer.

a $3.75 + 9.381 + 4.6 + 10.5$

b $14.807 + 6.6 - 7.22$

c 18.47×9.61

d 4.27×97.6

e $\dfrac{11.07+18.4}{12.2}$

f $\dfrac{38.18}{17.2-9.6}$

g $54.75 - 18.6 - 14.4$

h $\dfrac{18.46 \times 4.9}{39.72-15.2}$

i $62.13 \div 10.7$

j $(4.89)^2$

Did you know?

Degrees, minutes and seconds

We use base 10 systems in number, measurement and currency: there are 100 centimetres in a metre, 1000 grams in a kilogram and 100 cents in a dollar. So why are there 360° in a revolution, 60 minutes in a degree and 60 seconds in a minute?

In 2000 BCE, the Babylonians used a base 60 or sexagesimal system of numeration, because 60 is a rounder, more convenient number than 10. This is because 60 has more factors and is divisible by 3, 4 and 6. Furthermore, $6 \times 60 = 360$, which was the Babylonian approximation for the number of days in a year, so that each day the Earth would travel 1° around the Sun.

As measuring devices and calculations required greater precision, each degree was subdivided into 60 equal parts called minutes, and these were further divided into 60 parts called seconds. This level of accuracy is essential in navigation and mapping.

1 **A minute is a 'small' part of a degree. Investigate how an alternative meaning (and pronunciation) of 'minute' is 'tiny'.**

2 **A second is the 'second' subdivision of a degree. Explain how there are 2 different meanings of 'second'.**



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Finding an unknown side

4.05

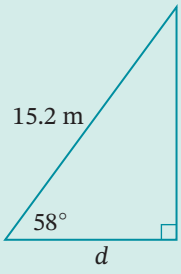
Since the trigonometric ratio of any angle is a constant number, we can use it to calculate the length of an unknown side in a right-angled triangle if one other side is known. We need to select the correct ratio that links the given angle to the unknown side and known side.

Videos
Finding an unknown side
Trigonometry

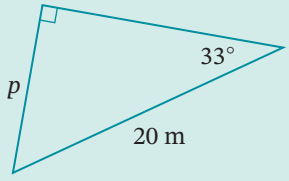
Example 11

Find the value of each variable, correct to 2 decimal places.

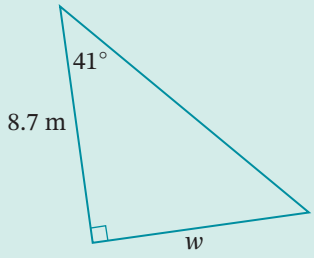
a



b



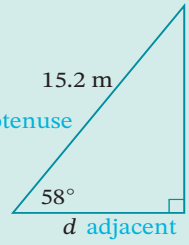
c



Solution

a SOH, CAH OR TOA?

The marked sides related to 58° are the adjacent side (A) and the hypotenuse (H), so use cos.



$$\cos 58° = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$= \frac{d}{15.2}$$

$$15.2 \times \cos 58° = 15.2 \times \frac{d}{15.2}$$ Multiplying both sides by 15.2

$$15.2 \cos 58° = d$$

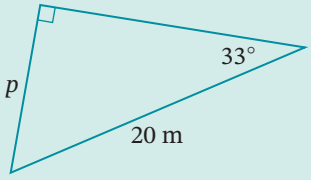
$$d = 15.2 \cos 58°$$

$$= 8.05477...$$

$$= 8.05$$

From the diagram, a length of 8.05 m looks reasonable.

b SOH, CAH OR TOA?



The marked sides related to 33° are the opposite side (O) and the hypotenuse (H), so use sin.

$$\sin 33° = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$= \frac{p}{20}$$

$$20 \times \sin 33° = 20 \times \frac{p}{20}$$

$$20 \sin 33° = d$$

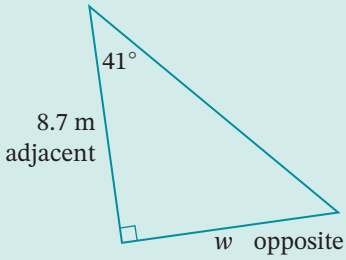
$$d = 20 \sin 33°$$

$$= 10.8927...$$

$$= 10.89$$

From the diagram, a length of 10.89 m looks reasonable.

c SOH, CAH OR TOA?



The marked sides related to 41° are the opposite side (O) and the adjacent side (A), so use tan.

$$\tan 41° = \frac{\text{opposite}}{\text{adjacent}}$$

$$= \frac{w}{8.7}$$

$$w = 8.7 \tan 41°$$

$$= 7.5627...$$

$$= 7.56$$

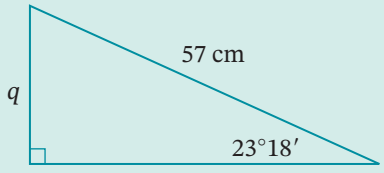
From the diagram, a length of 7.56 m looks reasonable.

Finding an unknown side in a right-angled triangle

- Identify the 2 labelled sides and decide whether to use sin, cos or tan
- Write an equation using the ratio, the given angle and the variable
- Solve the equation to find the value of the variable

Example 12

Find the value of q, correct to the nearest centimetre.



Solution

q is opposite 23° 18′, 57 cm is the hypotenuse, so use sin.

$$\sin 23° 18' = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\sin 23°18' = \frac{q}{57}$$

$$q = 57 \sin 23° 18'$$

$$= 22.5460...$$

$$= 23 \text{ cm}$$

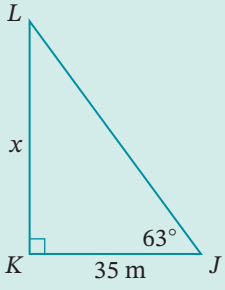
From the diagram, a length of 23 cm looks reasonable.

Example 13

$\triangle JKL$ is right-angled at K, $JK = 35$ m and $\angle J = 63°$. Find the length of LK, correct to the nearest metre.

Solution

Draw a diagram.



Let the length of LK be x.

x is opposite, 35 m is adjacent, so use tan.

$$\tan 63° = \frac{x}{35}$$

$$x = 35 \tan 63°$$

$$= 68.6913...$$

$$LK \approx 69 \text{ m}$$

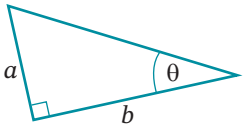
From the diagram, a length of 69 m looks reasonable.

EXERCISE 4.05 ANSWERS ON P. 571

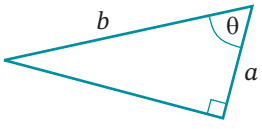
Finding an unknown side U F PS C

1 For each triangle, which trigonometric ratio (sin θ, cos θ or tan θ) is equal to $\frac{a}{b}$?

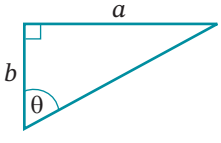
a



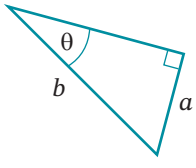
b



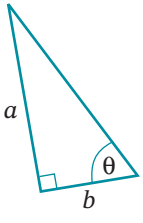
c



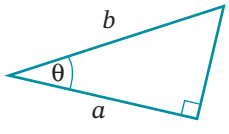
d



e

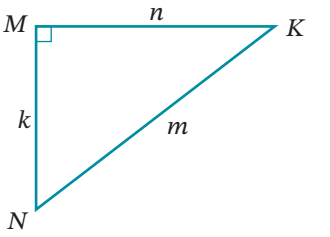


f



Foundation | Standard | Complex

2 For $\triangle KMN$, what is cos K? Select **A**, **B**, **C** or **D**.



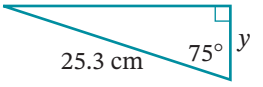
A $\frac{k}{m}$ **B** $\frac{m}{n}$

C $\frac{n}{m}$ **D** $\frac{k}{n}$

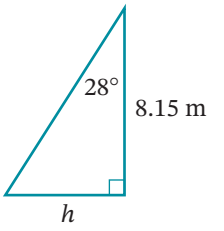
EXAMPLE 11

3 Find the value of the variable in each triangle, correct to 2 decimal places.

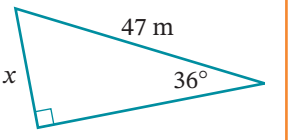
a



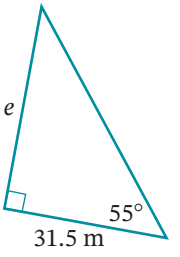
b



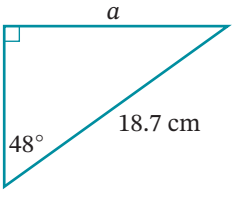
c



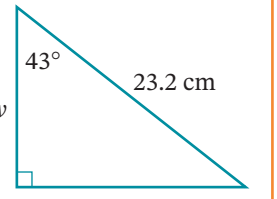
d



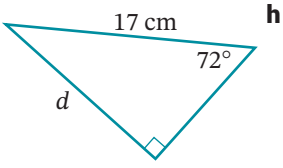
e



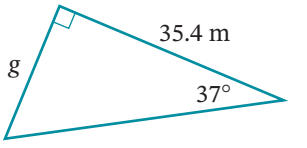
f



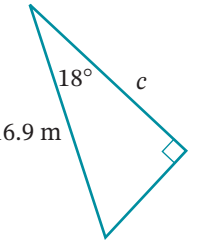
g



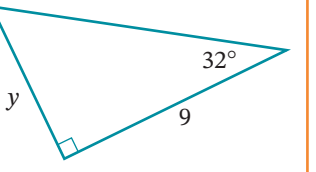
h



i



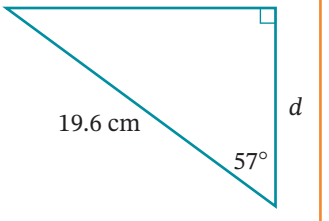
4 Which of the following is the correct expression for y in this triangle? Select **A**, **B**, **C** or **D**.



A 9 cos 32° **B** 9 tan 32°

C 9 sin 32° **D** 9 tan 58°

5 What is the length of d in this triangle? Select **A**, **B**, **C** or **D**.



A 10.67 cm **B** 10.68 cm

C 16.44 cm **D** 16.43 cm

Foundation Standard Complex

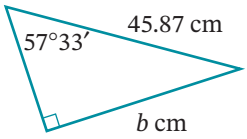
9780170479370

EXAMPLE 12

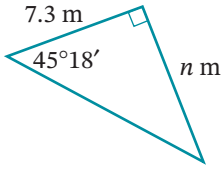
4.05

6 Find the value of the variable in each triangle, correct to 2 decimal places.

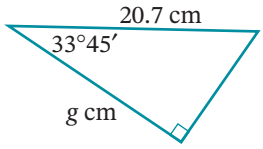
a



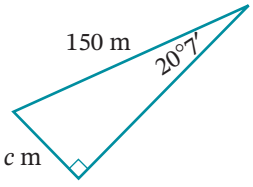
b



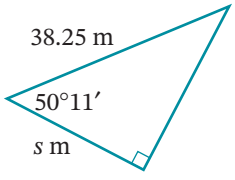
c



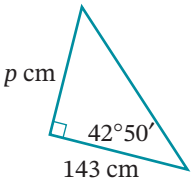
d



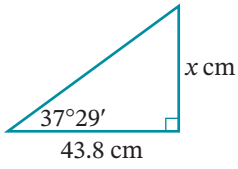
e



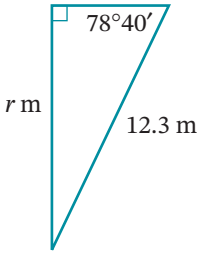
f



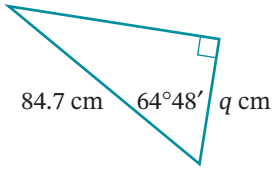
g



h



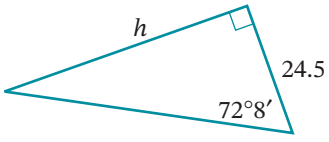
i



7 What is the value of h? Select **A**, **B**, **C** or **D**.

A 7.52 **B** 23.32

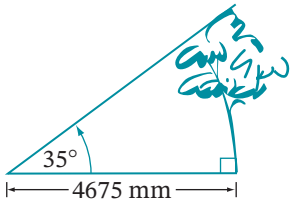
C 45.78 **D** 76.00



8 What is the height of this tree? Select **A**, **B**, **C** or **D**.

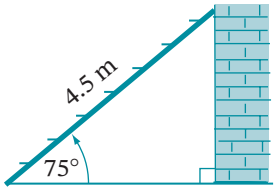
A 2.68 m **B** 3.27 m

C 3.83 m **D** 6.68 m

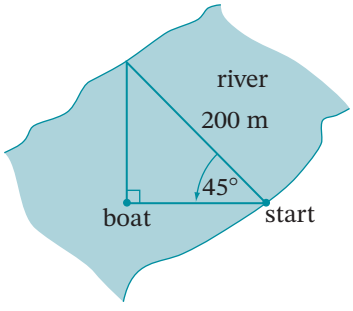


9 Find each length or distance, correct to one decimal place. PS

a How high the ladder reaches up the wall.

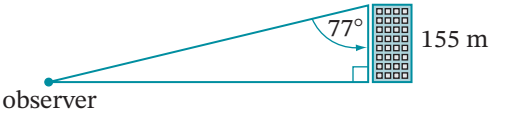


b The distance between the boat and the start.

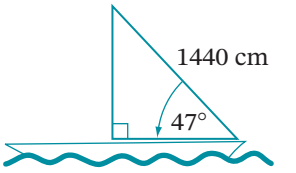


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c The distance from the observer to the base of the building.



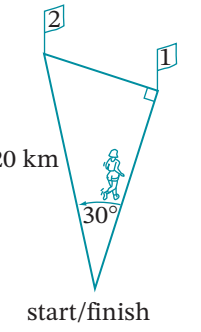
d The height of the boat's mast.



e The distance between:

i checkpoints 1 and 2

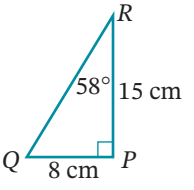
ii checkpoint 1 and the start.



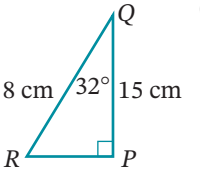
EXAMPLE 13

10 In $\triangle PQR$, $\angle P$ is right-angled, $\angle Q = 32°$, $PR = 8$ cm and $QR = 15$ cm. Which diagram fits this description? Select **A**, **B**, **C** or **D**. **C**

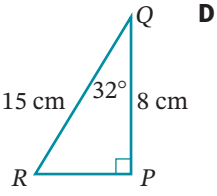
A



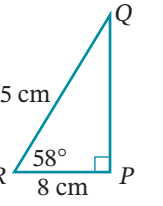
B



C



D



11 $\triangle ABC$ is right-angled at B, $AC = 14.8$ m and $\angle C = 56°$. Find the length of side AB, correct to one decimal place.

12 $\triangle MNR$ is right-angled at M, $MN = 19$ cm and $\angle N = 27°$. Find the length of MR, correct to the nearest centimetre.

13 In $\triangle XYW$, $\angle X = 90°$, $\angle Y = 43.7°$ and $WY = 8.34$ m. Find the length of XW, correct to 2 decimal places.

14 $\triangle AHK$ is right-angled at K, $\angle H = 76°$ and $AH = 13.9$ m. Find the length of HK, correct to one decimal place.

15 A tree casts a shadow 20 m long. If the Sun's rays meet the ground at 25°, find the height of the tree, correct to the nearest cm. **PS**

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16 A 6 m ladder is placed against a pole. If the ladder makes an angle of 17° with the pole, how far up the pole does the ladder reach? Answer to the nearest mm. PS

17 A golfer is 180 m in a straight line from the 8th hole. The ball is hit 15° to the right of the hole but still ends up level with the hole. How far is the ball from the hole? Answer to the nearest metre. PS

18 A park is in the shape of a rectangle. A path 450 m in length crosses the park diagonally. If the path makes an angle of 36° with the longer side, find the dimensions of the park. Answer to the nearest metre. PS

19 A wheelchair ramp is 6 m long and makes an angle of 4.5° with the ground. How high is the top of the ramp above the ground (correct to 2 decimal places)? PS

20 A boat is anchored by a rope 5.5 m long. If the anchor rope makes an angle of 23° with the vertical, calculate the depth of the water (correct to one decimal place). PS

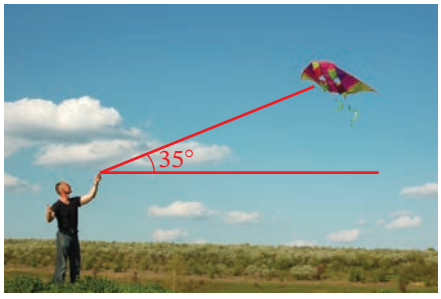
21 A rectangular gate has a diagonal brace that makes an angle of 60° with the bottom of the gate. If the length of the diagonal brace is 1860 mm, calculate the height of the gate. Select **A**, **B**, **C** or **D**. PS

A 2148 mm **B** 930 mm **C** 1611 mm **D** 3221 mm

22 Tayyab is flying a kite that is attached to a string 155 m long. The string makes an angle of 35° to the horizontal. How high, to the nearest metre, is the kite above Tayyab's hand? Select **A**, **B**, **C** or **D**. PS

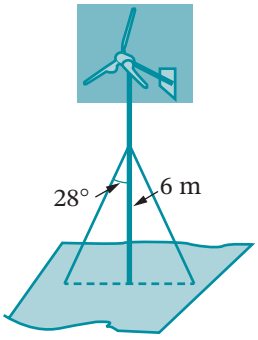
A 78 m **B** 89 m

C 109 m **D** 127 m



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23 A wind turbine is supported by 2 wires that are attached to the pole 6 m above the ground. The wires make an angle of 28° with the pole. Calculate the distance from the bottom of the wire to the base of the pole, correct to the nearest centimetre. PS



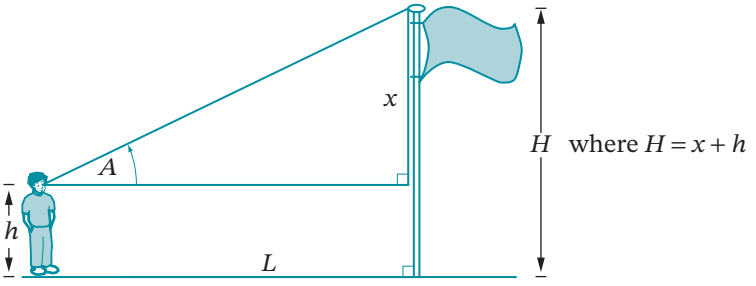
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Investigation

Calculating the height of an object

You will need: tape measure or trundle wheel, a clinometer (or protractor) to measure the angle.

Trigonometry can be used to find the heights of buildings, flagpoles and trees without actually measuring them. This can be done by measuring the distance along the ground from the base of the object to a person. The person then measures the angle to the top of the object. For example, the height of a flagpole can be calculated using the set-up shown in the diagram.



- h is the eye height of the person who measures the angle, A, to the top of the flagpole
- L is the distance the person is from the base of the flagpole
- x is the height of the flagpole above the person's eye height
- $H = x + h$ is the height of the flagpole above the ground.

1 Select a tall object outside to measure.

2 Work with a partner to measure (in cm) the distance, L, along the ground, the height, h, of the person, and the angle (in degrees) to the top of the object. Copy the table below and record your information in the first row.

Distance, L (cm)	Angle, $A°$	Height of person, h (cm)	Calculated height, x cm	Height of flagpole, H cm

3 Use the tan ratio to calculate the value of x, correct to the nearest whole number.

4 Hence find H, the height of the kite, correct to the nearest centimetre. Write your answers in the table.

5 Repeat the measurements and calculations three more times from different positions, with different persons measuring the angle. This will help to improve the accuracy of your results and minimise errors. Write your results in the table.

6 Did you find similar values for H? Do they seem reasonable for the height of the object?

7 Calculate the average value for H.

Finding more unknown sides

4.06

In the following examples, the unknown appears in the **denominator** of the equation.

Worksheet
Finding an unknown side

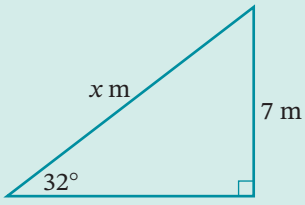
Puzzle
Trigonometry equations 1

Video
Finding the hypotenuse

Using sin or cos to find the hypotenuse

Example 14

Find the value of x, correct to one decimal place.



Solution

7 m is the opposite side, x m is the hypotenuse, so use sin.

$$\sin 32° = \frac{7}{x}$$

Note that the variable x appears in the denominator of the equation.

$$\sin 32° \times x = \frac{7}{x} \times x$$ Multiply both sides by x

$$x \sin 32° = 7$$

$$\frac{x \sin 32°}{\sin 32°} = \frac{7}{\sin 32°}$$ Divide both sides by sin 32°

$$x = \frac{7}{\sin 32°}$$

$$= 13.2095...$$

$$\approx 13.2$$

Note that when the unknown appears in the denominator of an equation, it can swap positions with the trigonometric ratio, so that $\sin 32° = \frac{7}{x}$ becomes $x = \frac{7}{\sin 32°}$.

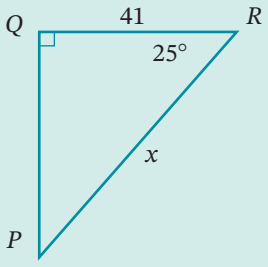
Example 15

$\triangle PQR$ is right-angled at Q, $QR = 41$ m and $R = 25°$. Find RP, correct to the nearest metre.

Solution

Let $x = RP$.

41 m is the adjacent side, x is the hypotenuse, so use cos.



$$\cos 25° = \frac{41}{x}$$

Note that the variable x appears in the denominator of the equation.

$$x = \frac{41}{\cos 25°}$$

Swap the position of x with cos 25°

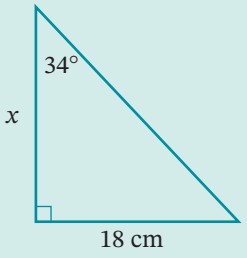
$$= 45.2384...$$

$$\approx 45 \text{ m}$$

Using tan to find the adjacent side

Example 16

Find the length of x, correct to 2 decimal places.



Solution

18 cm is the opposite side, x is the adjacent side, so use tan.

$$\tan 34° = \frac{18}{x}$$

x appears in the denominator

$$x = \frac{18}{\tan 34°}$$

Swap the position of x with tan 34°

$$= 26.6860...$$

$$\approx 26.69 \text{ cm}$$

Alternative method

To avoid having x in the denominator, we could use tan with the 3rd angle of the triangle.

3rd angle $= 180° - 90° - 34° = 56°$

$$\tan 56° = \frac{x}{18}$$

$$x = 18 \tan 56°$$

$$= 26.6860...$$

$$\approx 26.69 \text{ cm}$$

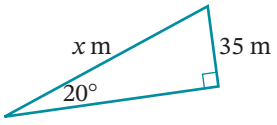
EXERCISE 4.06 ANSWERS ON P. 571

Finding more unknown sides U F PS

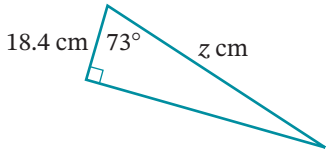
EXAMPLE 14

1 Find the value of each variable, correct to one decimal place.

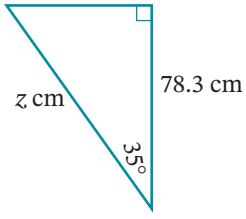
a



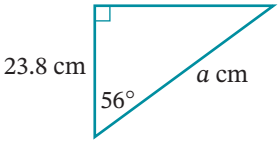
b



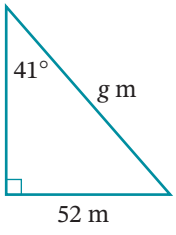
c



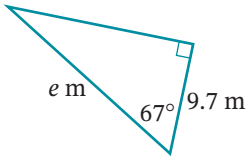
d



e

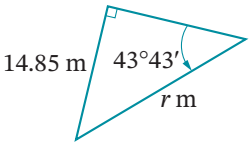


f

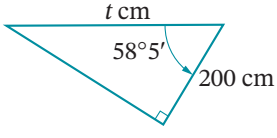


2 Find the value of each variable, correct to one decimal place.

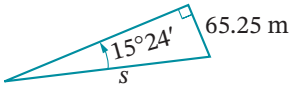
a



b



c



EXAMPLE 15

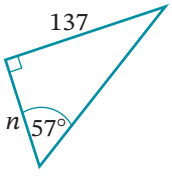
3 $\triangle XYZ$ is right-angled at Z, $ZY = 230$ mm and $\angle Y = 45°$. Find the length of XY, correct to the nearest millimetre.

4 In $\triangle KLW$, $\angle L = 90°$, $KL = 12$ m and $\angle W = 75.2°$. Find KW, correct to one decimal place.

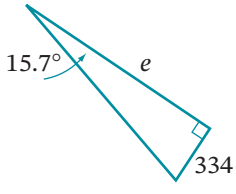
EXAMPLE 16

5 Find the value of each variable, correct to 2 decimal places.

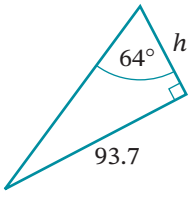
a



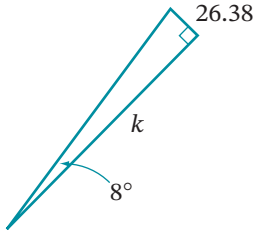
b



c



d

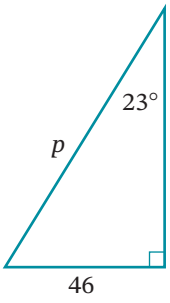


6 $\triangle CDE$ is right-angled at D, $\angle E = 36°$ and $CD = 5$ m. Find the length of side DE, correct to 2 decimal places.

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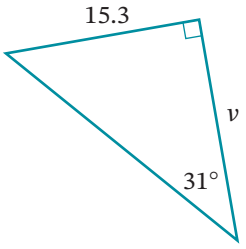
7 What is the correct expression for p in this triangle?
Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{46}{\cos 23°}$ **B** $\frac{46}{\sin 23°}$ **C** $46 \cos 23°$ **D** $46 \sin 23°$



8 What is the value of v in this triangle? Select **A**, **B**, **C** or **D**.

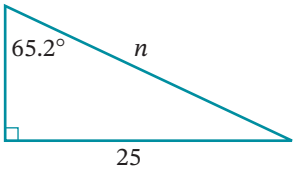
A 25.46 **B** 9.19 **C** 29.71 **D** 17.85



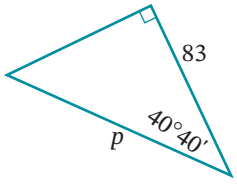
9 In $\triangle HMT$, $\angle T = 90°$, $\angle M = 19°\ 47'$ and side $HT = 18.4$ cm. Find the length of side HM, correct to one decimal place.

10 Find the value of each variable, correct to one decimal place.

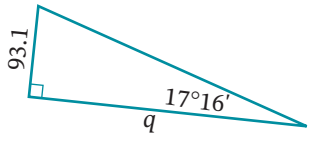
a



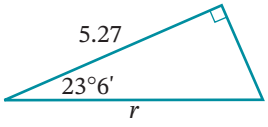
b



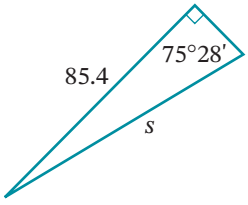
c



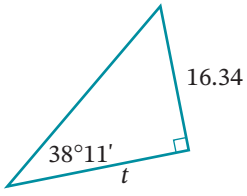
d



e

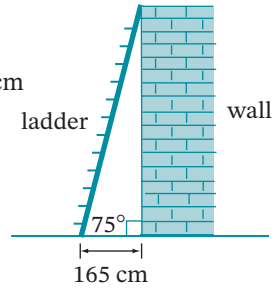


f



11 Find the length of this ladder. Select **A**, **B**, **C** or **D**.

A 159 cm **B** 171 cm **C** 243 cm **D** 638 cm

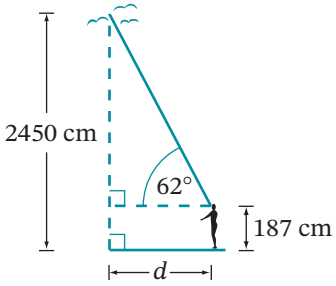


Foundation Standard Complex

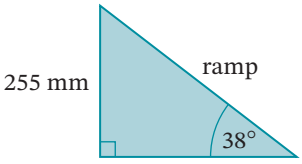
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12 Find each length or distance, correct to one decimal place. PS

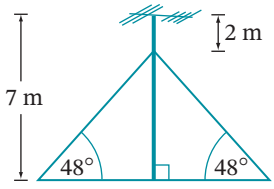
a How far the person is from being directly under the birds?



b The length of the ramp.



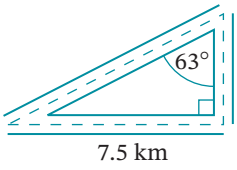
c The length of the support wire.



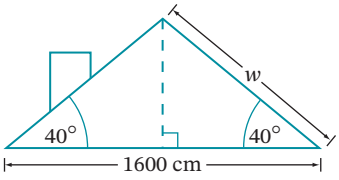
d The length of:

i the shortest road

ii the longest road.



e The slant height of the roof.



13 $\triangle FGW$ is right-angled at F, $\angle W = 84°$ and $WF = 42.1$ m. Find the length of WG, correct to one decimal place.

EXAMPLE 15

14 A ladder rests against a wall. The foot of the ladder is 355 cm from the wall and makes an angle of 63° with the ground. How long (to the nearest cm) is the ladder? PS

15 A supporting wire is attached to the top of a flagpole. The wire meets the ground at an angle of 51° and the flagpole is 15 m high. How far, to the nearest 0.1 m, from the base of the flagpole is the wire anchored to the ground? PS

16 A glider is flying at an altitude (height) of 1.5 km. To land, it descends at an angle of 18° to the ground. How far, to the nearest 0.1 km, must the glider travel before landing? PS



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17 The entrance to the school library is 60 cm above ground level. A wheelchair ramp is built to the entrance at an angle of 5° with the ground. How long (to the nearest 0.01 m) is the ramp? PS

18 A shooter aims directly at a target, but just before firing, the rifle is lifted 1° off target. The shot misses the target by 67 mm. How far is the shooter standing from the target? Select **A**, **B**, **C** or **D**. PS

A 1169 mm **B** 3838 mm **C** 3839 mm **D** 6701 mm

19 A hot air balloon is anchored to the ground by a rope. When it drifts 20 m sideways, it makes an angle of 75° with the ground. How long is the rope (to one decimal place)? PS

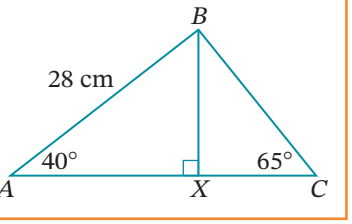


iStock.com/Asoa-Pacific Images Studio

20 For $\triangle ABC$, $\angle A = 40°$, $\angle C = 65°$, $AB = 28$ cm and BX is perpendicular to AC. Calculate, correct to one decimal place, the length of:

a BX

b CX



Investigation

Finding an angle, given a trigonometric ratio

You will need: a ruler, compasses and a calculator.

1 Copy and complete this table, calculating each ratio as a decimal, correct to 3 decimal places.

θ	sin θ	cos θ	tan θ
0°			
15°			
30°			
45°			
60°			
75°			
90°			undefined

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2 Can you work out why there is no answer for tan 90°?

3 What are the minimum and maximum values of sin θ?

4 What are the minimum and maximum values of cos θ?

5 Is there a pattern between the values of sin θ and cos θ?

6 Check the value of sin 30° by constructing a right-angled triangle with one angle that is 30°, measuring the opposite side and hypotenuse and dividing them.

7 Check the value of tan 45° by constructing a right-angled triangle with one angle that is 45°, measuring the opposite and adjacent sides and dividing them.

8 Use the table to estimate each trigonometric ratio and check your estimate using a calculator.

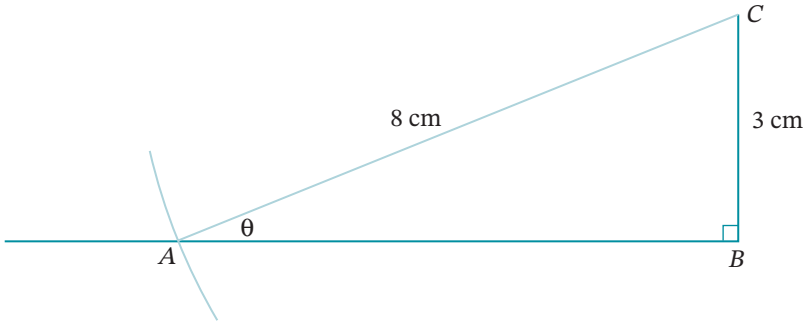
a sin 80° **b** cos 34° **c** tan 55°

9 If $\sin\theta = \frac{3}{8}$, find the value of the unknown angle θ, correct to the nearest degree:

a by using the table and estimating (change $\frac{3}{8}$ to a decimal first)

b using a calculator to guess-and-check

c constructing a right-angled triangle with one angle θ, opposite side 3 cm and hypotenuse 8 cm, then measuring the size of θ.



10 If $\cos\theta = \frac{2}{5}$, find the value of the unknown angle θ, correct to the nearest degree:

a by using the table and estimating (change $\frac{2}{5}$ to a decimal first)

b using a calculator to guess-and-check

c constructing a right-angled triangle with one angle θ, adjacent side 2 cm and hypotenuse 5 cm, then measuring the size of θ.

11 If $\tan\theta = \frac{7}{10}$, find the value of the unknown angle θ, correct to the nearest degree:

a by using the table and estimating (change $\frac{7}{10}$ to a decimal first)

b using a calculator to guess-and-check

c constructing a right-angled triangle with one angle θ, opposite side 7 cm and adjacent side 10 cm, then measuring the size of θ.

4.07 Finding an unknown angle

Worksheets
Finding an unknown angle
Trigonometry problems

Puzzles
Trigonometry squaresaw
Trigonometry: Finding angles
Trigonometry equations 2

Presentation
Applications of triangles

A scientific calculator can be used to evaluate a trigonometric ratio such as sin 38°, but it can also be used to find an unknown angle, θ, if the trigonometric ratio of the angle is known, for example, if sin θ = 0.9063.

An unknown angle can be found using the [sin^{-1}] [cos^{-1}] [tan^{-1}] keys on the calculator. These are called the **inverse sin**, **inverse cos** and **inverse tan** functions, found by pressing the [SHIFT] or [2nd F] key before the [sin], [cos] or [tan] key.

Example 17

a If sin θ = 0.9063, find angle θ, correct to the nearest degree.

b If tan X = 3.754, find angle X, correct to the nearest minute.

c If cos α = $\frac{4}{7}$, find angle α, correct to the nearest degree.

Solution

a
$\sin\theta = 0.9063$
$\theta = 64.9989...°$
$\approx 65°$

On a calculator: [SHIFT] [sin] 0.9063 [=]

b
$\tan X = 3.754$
$X = 75.0837...°$
$= 75° 5' 1.62''$
$\approx 75° 5'$

[SHIFT] [tan] 3.754 [=]
[°’’’] or [DMS]

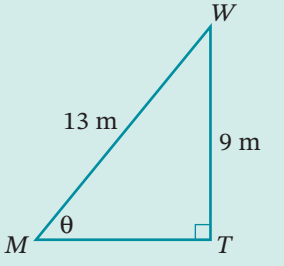
c
$\cos\alpha = \frac{4}{7}$
$\alpha = 55.1500...°$
≈ 55

[SHIFT] [cos] 4 [▭/▭] 7 [=]
Note: [▭/▭] or [a$^{b/c}$] is the fraction key.

Videos
Finding an unknown angle
Trigonometry

Example 18

Find the size of angle θ, correct to the nearest degree.



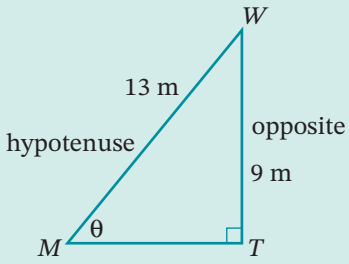
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Solution

SOH, CAH or TOA?

The known sides for θ are the opposite (O) side and the hypotenuse (H), so use sin.

$\sin\theta = \frac{9}{13}$



$\theta = 43.8130...°$

$\approx 44°$

From the diagram, an angle size of 44° looks reasonable.

SHIFT sin 9 ▭ 13 =

Finding an unknown angle in a right-angled triangle

- Identify the 2 known sides and decide whether to use the sin, cos or tan ratio.
- Write an equation using the ratio, the angle and the 2 sides as a fraction.
- Use the calculator's inverse trigonometric function to find the size of the angle.

Example 19

$\triangle XYZ$ is right-angled at Y, with $XY = 35$ cm and $YZ = 47$ cm. Find $\angle Z$, correct to the nearest degree.

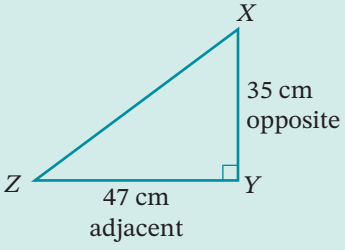
Solution

Sketch a diagram.

SOH, CAH or TOA?

The known sides to $\angle Z$ are the opposite (O) and the adjacent (A), so use tan.

$\tan Z = \frac{35}{47}$



$Z = 36.6743...°$

$\approx 37°$

From the diagram, an angle size of 37° looks reasonable.

SHIFT tan 35 ▭ 47 =

EXERCISE 4.07 ANSWERS ON P. 572

Finding an unknown angle U F PS

EXAMPLE 17

1 Find the size of angle θ, correct to the nearest degree.

a $\cos\theta = 0.76$ **b** $\tan\theta = 2.0532$ **c** $\sin\theta = \frac{\sqrt{3}}{2}$

d $\tan\theta = 6$ **e** $\sin\theta = \frac{7}{8}$ **f** $\cos\theta = \frac{13}{15}$

g $\sin\theta = \frac{1}{10}$ **h** $\cos\theta = \frac{1}{\sqrt{2}}$ **i** $\tan\theta = \sqrt{3}$

j $\cos\theta = 0.1352$ **k** $\tan\theta = 8.836$ **l** $\sin\theta = \frac{1}{4}$

2 Find the size of angle A, correct to the nearest minute.

a $\tan A = \frac{15}{7}$ **b** $\sin A = 0.815$ **c** $\cos A = \frac{4}{5}$

d $\cos A = 0.9387$ **e** $\tan A = \frac{19}{20}$ **f** $\cos A = \frac{3}{10}$

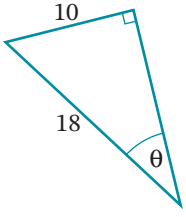
g $\sin A = \frac{5}{11}$ **h** $\sin A = 0.88$ **i** $\tan A = 15.07$

j $\cos A = \frac{1}{7}$ **k** $\tan A = \frac{\sqrt{2}}{2}$ **l** $\sin A = \frac{7}{9}$

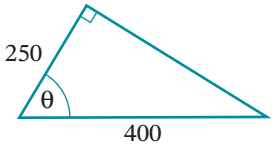
EXAMPLE 18

3 Find θ, correct to the nearest degree.

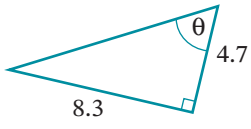
a



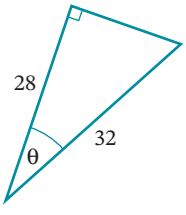
b



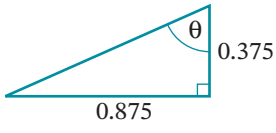
c



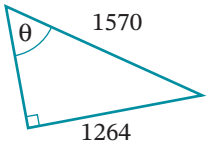
d



e



f

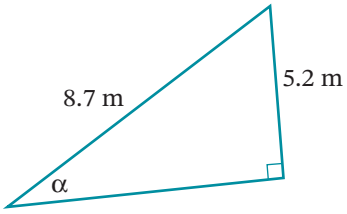


Foundation Standard Complex

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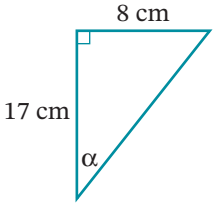
4 Find the size of angle α, correct to the nearest minute.
Select the correct answer **A**, **B**, **C** or **D**.

A 30° 52′ **B** 30° 53′
C 36° 42′ **D** 36° 43′

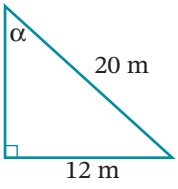


5 Find the size of angle α, correct to the nearest minute.

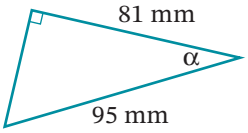
a



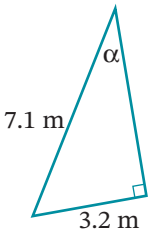
b



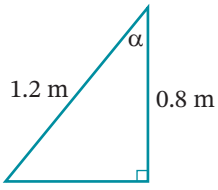
c



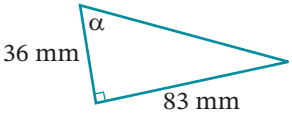
d



e

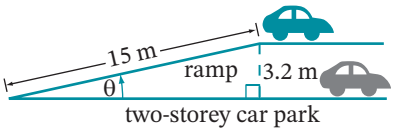


f

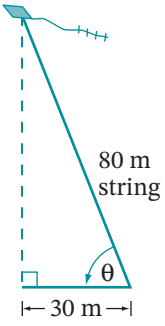


6 Find the size of angle θ, correct to the nearest degree.

a

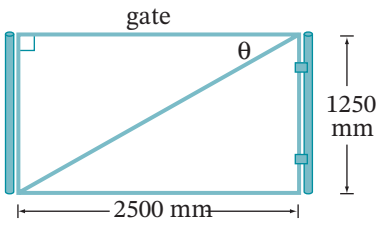


b

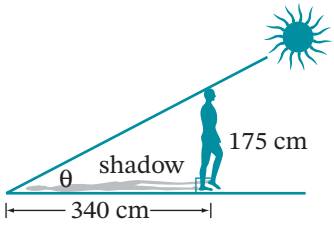


Foundation Standard Complex

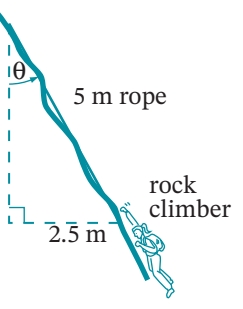
c



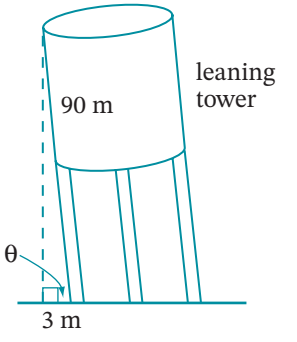
d



e



f



EXAMPLE 19

7 In $\triangle XYW$, $\angle X = 90°$, $XY = 8$ cm and $XW = 10$ cm. Find $\angle W$, correct to the nearest degree.

8 $\triangle HTM$ is right-angled at T, $HM = 45$ m and $MT = 35$ m. Find $\angle M$, correct to one decimal place.

9 In $\triangle FGH$, $\angle G = 90°$, $GH = 3.7$ m and $FH = 19.5$ m. Find the size of angle F, correct to the nearest minute.

10 In $\triangle CDE$, $\angle C = 90°$, $CD = 54$ mm and $CE = 42$ mm. Find $\angle E$, correct to the nearest minute.

For questions 11 to 17, write answers correct to the nearest degree.

11 A stretch of freeway rises 55 m for every 300 m travelled along the road. Find the angle at which the road is inclined to the horizontal. PS

12 A ladder 20 m long is placed against a building. If the ladder reaches 16 m up the building, find the ladder's angle of inclination to the building. PS

13 An aircraft is descending in a straight line to an airport. At a height of 1270 m, it is 1500 m horizontally from the airport. Find its angle of descent to the horizontal. Select **A**, **B**, **C** or **D**. PS

A 32° **B** 40° **C** 50° **D** 58°

14 A tree 8.5 m high casts a shadow 3 m long. What is the angle of the Sun from the ground? PS

15 At a resort, an artificial beach slopes down at a steady angle. After walking 8.5 m down the slope from the water's edge, the water has a depth of 1.6 m. At what angle is the beach inclined to the horizontal? PS

Foundation Standard Complex

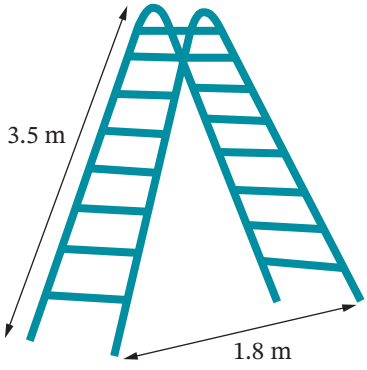
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16 A pile of wheat is in the shape of a cone that has a diameter of 35 m and measures 27 m up the slope to the apex. Calculate the angle of repose of the wheat (the angle the sloping side makes with the horizontal base). PS

17 A ship is anchored in water 40 m deep by a 65 m anchor chain. Find the angle at which the chain is inclined to the sea floor. PS

18 The arms of this fold out ladder are 3.5 m long. When fully spread, the feet of the ladder are 1.8 m apart.

Calculate, correct to the nearest minute, the angle between the arms of the ladder when fully spread. PS



Power plus ANSWERS ON P. 572

1 **a** Copy and complete each pair of trigonometric ratios, correct to 3 decimal places.

i $\sin 20° =$ _____, $\cos 70° =$ _____ **ii** $\sin 47° =$ _____, $\cos 43° =$ _____

iii $\sin 55° =$ _____, $\cos 35° =$ _____ **iv** $\sin 85° =$ _____, $\cos 5° =$ _____

b What do you notice about each pair of answers in part **a**?

c What do you notice about each pair of angles in part **a**?

d If $\cos 30° \approx 0.8660$ and $\sin \theta \approx 0.8660$, what is the value of θ?

e Copy and complete each equation.

i $\sin 75° = \cos$ _____ **ii** _____ $80° = \cos 10°$

iii $\cos$ _____ $= \sin 72°$ **iv** $\sin 30° =$ _____ $60°$

v $\cos 65° = \sin$ _____ **vi** $\sin$ _____ $= \cos 58°$

f Copy and complete this general rule: $\sin x = \cos$ (_____).

g Use a right-angled triangle with one angle x and sides a, b and c to prove that the above rule is true.

2 A plane is flying at an angle of 15° inclined to the horizontal.

a How far will the plane have to travel along its line of flight to increase its altitude (height) by 500 m?

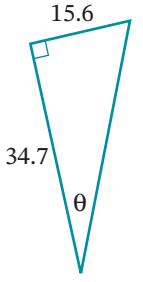
b At what angle must the plane climb to achieve an increase in altitude of 500 m in half the distance needed at an angle of 15°?

3 If $\sin 30° = \frac{1}{2}$, find, as a surd, the value of:

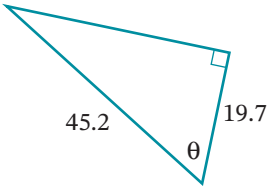
a $\cos 30°$ **b** $\tan 30°$

4 Find angle θ, correct to the nearest second.

a



b



5 By drawing an appropriate triangle, prove that:

a $\tan 45° = 1$ **b** $\sin 45° = \frac{1}{\sqrt{2}}$ **c** $\cos 45° = \frac{1}{\sqrt{2}}$

6 Sketch a right-angled triangle with side lengths a, b and c, where c is the hypotenuse. Let θ be one of the acute angles. Prove that $(\sin \theta)^2 + (\cos \theta)^2 = 1$.

CHAPTER 4 REVIEW

Quiz
Language of maths 4

Puzzle
Trigonometry crossword

Language of maths

adjacent	alpha (α)	clinometer	cosine (cos)
degree (°)	denominator	horizontal	hypotenuse
inverse ($^{-1}$)	minute (′)	mnemonic	opposite
phi (φ)	Pythagoras' theorem	right-angled	second (″)
similar triangles	sine (sin)	tangent (tan)	theta (θ)
trigonometry	trigonometric ratio	unknown	vertical

1 When measuring angle size, what is a second and what is its symbol?

2 What word means 'next to'?

3 Which side of a right-angled triangle is fixed and does not depend on the position of an angle?

4 What are the first 2 letters of the Greek alphabet?

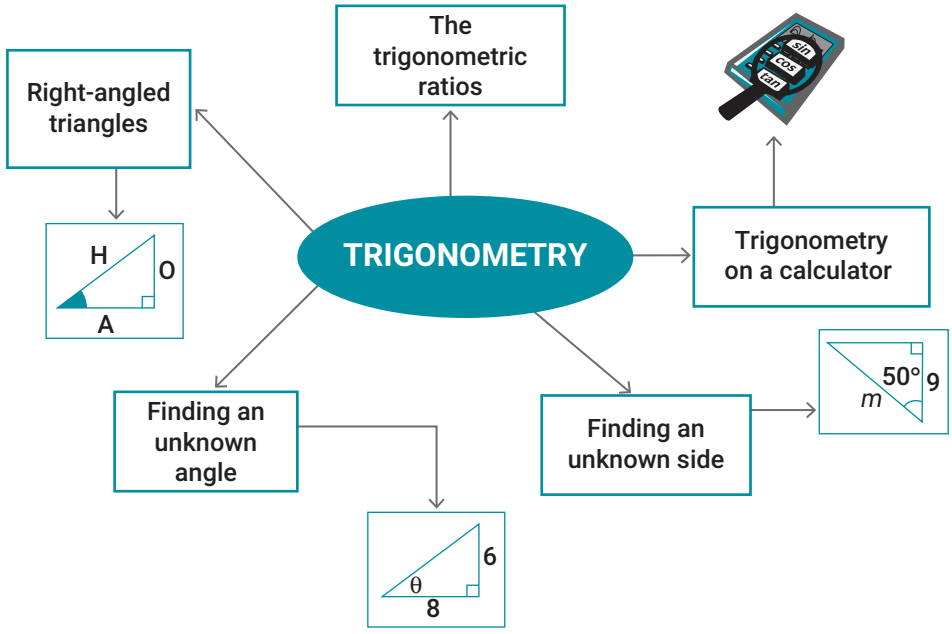
5 The word 'minute' has an alternative pronunciation and meaning. What is its alternative meaning?

6 What does 'inverse' mean and how is it used in trigonometry?

Topic summary

Worksheet
Mind map: Trigonometry

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



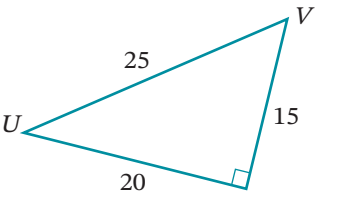
Quiz
Test yourself 4

TEST YOURSELF 4

ANSWERS ON P. 572

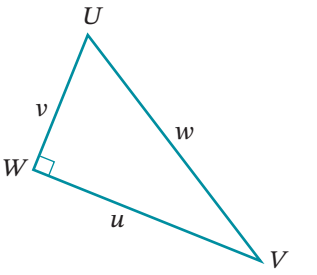
4.01

1 For angle U, name the opposite and adjacent sides and the hypotenuse.



4.01

2 For angle V, name the opposite and adjacent sides and the hypotenuse.

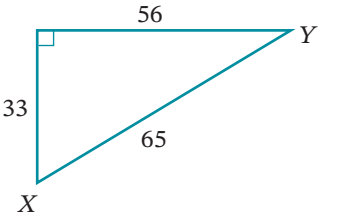


4.02

3 For this triangle, write as a fraction:

a $\sin Y$ **b** $\tan Y$

c $\sin X$ **d** $\cos X$



4.02

4 If $\sin \alpha = \frac{36}{85}$, find the values of $\cos \alpha$ and $\tan \alpha$ as fractions. (Draw a diagram.)

4.03

5 Construct a large right-angled triangle with an angle of 42°, then by measurement and calculation, find the value of each trigonometric ratio, correct to 3 decimal places.

a tan 42° **b** cos 42° **c** sin 42°

Compare your answers to the calculator's answers.

4.04

6 Round each angle to the nearest degree.

a 64° 27′ **b** 25° 43′ **c** 12° 8′ 50″

4.04

7 Round each angle to the nearest minute.

a 50° 19′ 26″ **b** 31° 55′ 55″ **c** 64° 18′ 30″

4.04

8 Evaluate each expression, correct to 4 decimal places.

a cos 32° **b** sin 50° 9′ **c** tan 8° 45′

d 200 tan 18° **e** 14 sin 87° 40′ **f** $\frac{13}{\cos 18°27'}$

4.04

9 Convert each angle size to degrees and minutes, correct to the nearest minute.

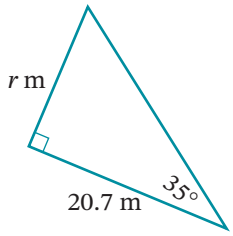
a 45.8° **b** 33.175° **c** 5.346°

Foundation Standard Complex

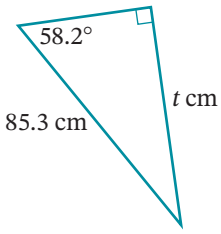
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10 Find the value of each variable, correct to 2 decimal places. 4.05

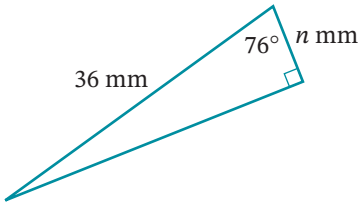
a



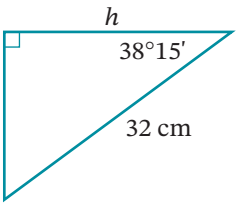
b



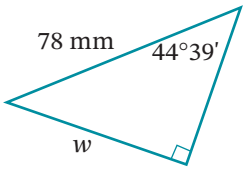
c



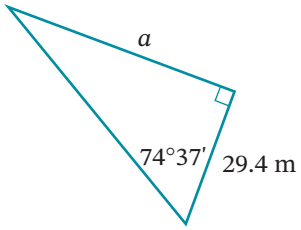
d



e



f

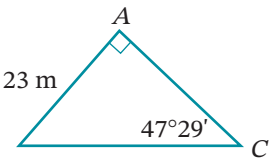


11 For each triangle, find the length of side AC, correct to one decimal place. 4.06

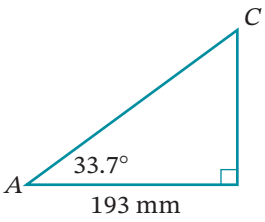
a



b



c

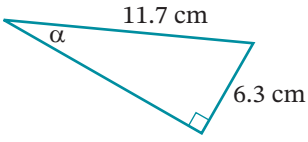


12 Find θ, correct to the nearest degree. 4.07

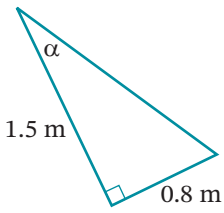
a $\tan\theta = 2.57$ **b** $\cos\theta = \frac{4}{7}$ **c** $\sin\theta = \frac{1.5}{1.6}$

13 Find the size of angle α, correct to the nearest degree. 4.07

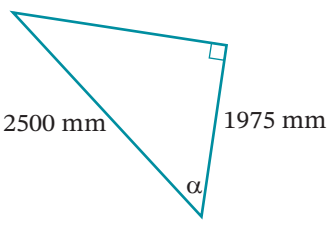
a



b



c



14 In $\triangle AEC$, $\angle C = 90°$, $CE = 3.9$ m and $AE = 4.2$ m. Find $\angle A$, correct to the nearest minute. 4.07

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